CLCSA16 DWDM

CLASSROOM ACTIVITY

SHORT ANSWER TEST

TOPIC: Classification of Data Mining Systems

Expanded Answers

1. Define a Data Mining System.

Answer: A Data Mining System is a computer-based system used to discover useful information, hidden patterns, relationships, and trends from large amounts of data. It combines database technology, statistics, machine learning, and artificial intelligence to analyze data and support decision-making. The main activities include collecting data, cleaning and preparing it, selecting relevant information, applying mining techniques, evaluating the results, and presenting useful knowledge. Example: An online shopping website analyzes customers' purchase histories to recommend products they may like.

2. List the different classifications of Data Mining Systems.

Answer: Data Mining Systems are commonly classified according to: (1) the type of data mined, such as relational databases, text, images, spatial data, and web data; (2) the type of knowledge mined, such as classification, clustering, association rules, prediction, and outlier detection; (3) the techniques used, such as statistics and machine learning; (4) the application domain, such as healthcare, banking, retail, and education; and (5) the degree of user interaction, such as interactive or automated systems. These classifications help identify the most suitable system for a particular problem.

3. Explain the classification of Data Mining Systems based on the type of data mined.

Answer: Based on the type of data, data mining systems include: (1) Relational database mining, which discovers patterns in tables; (2) Data warehouse mining, which analyzes integrated historical data; (3) Transactional data mining, which studies purchases or other transactions; (4) Spatial data mining, which finds patterns based on location; (5) Temporal and time-series mining, which analyzes changes over time; (6) Text mining, which extracts information from documents; (7) Multimedia mining, which analyzes images, audio, and video; (8) Web mining, which studies web pages, links, and user behaviour; and (9) Data-stream mining, which processes continuously arriving data. For example, a map service may use spatial mining to identify areas with heavy traffic.

4. Describe the classification of Data Mining Systems based on the type of knowledge mined.

Answer: Systems can be classified by the kind of knowledge they discover. Characterization summarizes the general features of a group, while discrimination compares one group with another. Association and correlation analysis identify relationships between items, such as products often purchased together. Classification assigns records to predefined categories, and prediction estimates unknown or future values. Clustering groups similar records without predefined labels. Outlier analysis detects unusual records, such as suspicious transactions. Evolution and trend analysis identify changes over time. These methods are selected according to the knowledge required.

5. Identify the techniques used in Data Mining Systems.

Answer: Common data-mining techniques include: (1) Classification, which assigns data to known categories; (2) Clustering, which groups similar records; (3) Association-rule mining, which finds items that occur together; (4) Regression, which estimates numerical values; (5) Prediction, which forecasts likely outcomes; (6) Outlier detection, which identifies unusual data; (7) Decision trees and neural networks, which learn patterns for prediction or classification; (8) Statistical analysis, which identifies relationships and summarizes data; and (9) Data visualization, which presents patterns through charts and graphs. The technique depends on the data and the goal of the analysis.

6. Explain the classification of Data Mining Systems based on applications.

Answer: Based on the application domain, data mining systems are used in several fields. In banking, they help detect fraud and assess credit risk. In retail, they analyze buying habits and recommend products. In healthcare, they support disease-risk analysis and patient-care planning. In telecommunications, they identify unusual calling patterns and predict customer churn. In education, they analyze student performance and identify learners who may need support. In manufacturing, they help predict equipment failure and improve quality. These systems turn domain-specific data into useful information for planning and decision-making.

7. Describe the classification of Data Mining Systems based on user interaction.

Answer: Based on user interaction, systems may be interactive or query-driven, semi-automated, or automated. Interactive systems allow users to ask questions, choose parameters, and explore results. Semi-automated systems perform much of the analysis but allow users to guide important steps. Automated systems run predefined mining processes with little user involvement and may generate alerts or reports automatically. Visual data-mining systems use charts and dashboards to help users understand results. For example, a business analyst may use a dashboard to filter sales data by region and compare trends.

8. Illustrate a real-world application of a Data Mining System in healthcare.

Answer: A hospital can use a data-mining system to analyze patients' age, symptoms, medical history, laboratory results, and lifestyle information to identify patterns linked to conditions such as diabetes or heart disease. The process involves collecting and cleaning records, selecting relevant features, applying classification or prediction models, and evaluating the results. The system can help healthcare professionals identify high-risk patients earlier, plan follow-up tests, and allocate resources. For example, a model may flag a patient for further assessment based on risk factors. Data mining supports clinical judgement; it does not replace doctors, and patient privacy and data security must be protected.

9. Define classification in data mining.

Answer: Classification is a supervised data-mining and machine-learning technique that learns from a labelled training dataset and uses the learned patterns to assign new records to predefined classes. The process generally involves preparing labelled data, training a model, testing its performance, and classifying new records. Common classification algorithms include decision trees, Naive Bayes, k-nearest neighbours, and support vector machines. For example, an email system classifies messages as 'spam' or 'not spam'. Classification is useful when the possible categories are already known.

10. Differentiate between classification and clustering in data mining.

Answer: Classification and clustering are both used to organize data, but they work differently. Classification is a supervised method that learns from labelled examples and assigns new records to predefined classes. Clustering is an unsupervised method that uses unlabelled data to group records based on similarity. Classification is commonly used for spam detection, disease-category prediction, and fraud detection. Clustering is used for customer segmentation, grouping similar documents, and discovering natural patterns. In short, classification predicts a known category, whereas clustering discovers groups that may not have been known in advance.